Observational Indistinguishability and Integrity Blind Regions Across the Evidence Boundaries of Hybrid Quantum-Classical Kernel Workflows

Roberto Fernández-Barrios, Iker Pastor-López, Amaia Pikatza-Huerga, and Pablo García Bringas

Abstract—Integrity auditing asks whether the evidence exposed at a workflow boundary can distinguish a consequential intervention from the baseline. We formalize this question for hybrid quantum-classical workflows as observational indistinguishability under a declared information set and trusted references. A structural blind region is the set of interventions whose view equals the baseline's; it is monotone under refinement and closed exactly by evidence that separates the intervention class. The evaluation-label path is a protected boundary: for a fixed predictor, label-only changes leave feature and prediction evidence invariant, and every change of the reported balanced accuracy changes the confusion matrix, so a trusted aggregate of the same batch certifies the conclusion, an item-aligned reference certifies item identity, and a statistical baseline yields only calibrated power. The blind regions are confirmed on 3,600 label interventions over CICIDS2017, UNSW-NB15 and ToN-IoT in eight fixed environments. A conformal family rule has finite-sample level at most α under exchangeability; in the executed, non-exchangeable design the observed false-alarm rates are 0.048–0.058. In an offline `allow/hold/block` evaluation over 24,000 frozen observations, batch-level regimes with feature evidence serve 4,365–4,496 of 7,008 materially changed results; the trusted-reference regime serves none but interrupts 629 of 1,200 near-null synthetic controls, mostly by statistical holds. An adaptive attacker who preserves the feature clusters that batch-level sensors rely on cuts drift detection from 0.96–1.00 to 0.01–0.66 while keeping 83–91% of the conclusion changes. A 165-cell simulator gate maps the lattice onto circuit, kernel and estimation evidence; a four-contract prototype executes the composed decision.

Index Terms—hybrid quantum-classical systems, integrity auditing, observability, quantum kernels, runtime assurance

1 Introduction

Hybrid quantum-classical software distributes one reported result across heterogeneous computational and evidential boundaries: classical records are acquired, cleaned, projected and encoded; a circuit and an execution stack produce quantum evidence; a classical learner turns a kernel into a decision; and an evaluation layer aligns that decision with reference outcomes and reports a metric. A plausible final number may therefore coexist with a failure in the data, circuit, kernel, prediction, label or reporting path, and security analyses of hybrid systems call for lifecycle-aware controls rather than trust in an isolated algorithm [1], [2].

Robustness benchmarking collapses this chain into a performance change, an endpoint that cannot say what changed or whether the available evidence could have revealed it. Consider the evaluation boundary of a hybrid security service that scores each batch with a fixed predictor and joins the predictions with a ground-truth feed from a separate subsystem (analyst verdicts, incident tickets, a delayed label store). The predictions are intact; the join or the store is stale, corrupted or rewritten by whoever can write to it, and balanced accuracy moves although the model has not. Feature and prediction monitors cannot observe the event because their inputs are identical; label counts expose an unbalanced corruption but not a balanced exchange of class identities; a signature on the label file authenticates the signed object, not the item-level correspondence between labels and the predictions being scored, so a correctly signed but stale or misaligned file passes. Calling the event stealthy omits the decisive qualifier: stealthy to which auditor, holding which evidence, trusting which reference? A trusted aggregate of the same batch certifies that the reported conclusion is the one the honest labels would give, only item alignment certifies that each label belongs to its item, and if one party controls both the label store and the reference used to check it, no local verifier can tell the substitution from an honest run (Proposition 6). The scenario is a reading aid, not an incidence estimate.

That qualifier has a long history outside machine learning: control theory characterizes stealthy attacks as inputs indistinguishable from nominal behaviour through the monitor's outputs [3]–[5], quantifies the detectability–impact trade-off [6], [7] and treats observability as a property of the monitored system [8]. This article transports that discipline to the evidence chain of a hybrid learning workflow: it treats the *evaluation-label path* as a protected boundary with its own blind regions; it separates structural indistinguishability (identical view) from statistical undetectability (a view that differs, but not beyond the null variability of fresh batches) and shows which *trusted same-batch reference* converts the second into an exact test; and it composes the coverage into an executable, calibrated decision whose costs are measured on both sides.

- *The authors are with the Faculty of Engineering, University of Deusto, Avda. de las Universidades 24, 48007 Bilbao, Spain. E-mail: roberto.fernandez.b@deusto.es (corresponding author), iker.pastor@deusto.es, a.pikatza@deusto.es, pablo.garcia.bringas@deusto.es.*
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The contribution is one package:

- an observation model of hybrid workflows with primitive and derived artifacts, in which auditability is a property of (intervention class, information set, trusted references), with monotonicity, closure and materiality results, minimal counterexamples, and the quantum branch as an instance of the same lattice with semantic, estimated and observed kernels kept apart;
- an explicit adversary and failure model for every executed intervention, with the roots it cannot write, including an executed adaptive attacker who preserves the fingerprint that batch-level sensors rely on;
- multi-dataset, fixed-OOD validation with verified deduplication, cluster-level uncertainty and preregistered null calibration;
- decision-level calibration by a conformal family rule whose finite-sample level holds under exchangeability, and an offline end-to-end `allow/hold/block` evaluation on 24,000 frozen observations whose endpoints are the materially changed results a policy serves and its cost on near-null variation; and
- a bounded quantum-specific layer and a four-contract prototype that execute the composed decision locally.

No constituent is claimed as individually sufficient novelty: the propositions are finite-batch statements, the family rule is a standard conformal construction, contracts, hash chains and semantics-preserving tests have clear prior art, and the secondary quantum-versus-classical impact profile (supplement) is neither a quantum-advantage result nor a vulnerability ordering. Physical hardware, calibrated backend noise, scheduling, provider-side security, multi-tenant interference, deployed runtime services and operational Fleet Management are outside the claim.

2 Related Work and Positioning

2.1 Stealth, Detectability and Observability in Secure Systems

The idea that detectability depends on what the monitor can see is established. Pasqualetti *et al.* characterize undetectable and unidentifiable attacks on cyber-physical systems through the observability of the monitored dynamics [3]; Teixeira *et al.* derive zero-dynamics and replay attacks that are stealthy to residual detectors [4]; Liu *et al.* exhibit false-data injections that lie in the estimator's range space [5]; later work quantifies how much an ε-stealthy attacker can degrade performance [6], [7] and surveys physics-based detection as a question of which residuals a monitor can form [8]. Runtime assurance routes a decision through a verified simple controller when a monitor flags the complex one [9], runtime verification checks executions against specifications [10], and integrity checkers compare stored digests against trusted baselines [11]. This article does not claim the general insight that detectability is information-relative; it differs in the object: an evidence chain whose protected boundaries include the ground truth used to score the model, blind regions derived from views rather than linear dynamics, the trusted reference made explicit as the difference between statistical and exact separation, and a calibrated decision evaluated end to end on frozen outputs, including against an adaptive attacker.

2.2 Monitoring, Labels and Evaluation Integrity

Label shift estimation targets a changed class prior under source-target assumptions [12]; MLDemon and uncertainty-aware monitoring combine unlabeled observations with selectively acquired labels [13], [14]; an evidence-sufficiency framework maps proxy coverage under delayed ground truth and formalizes the inability of feature proxies to expose concept drift [15]. Northcutt *et al.* document pervasive label errors in test sets [16]; Arp *et al.* catalogue sampling, labeling, metric and threat-model pitfalls in security benchmarks [17]; Bajaj names *evaluation blindness*, a measurement indistinguishable from a healthy state while the system fails [18]. These works treat labels as available, noisy or late; here the label path is an asset under intervention, availability is separated from trust, and the evidence needed to identify a corruption is derived, not assumed. VAMP [19] extends media-provenance manifests to machine learning: datasets, software, models and evaluation sets are signed and bound by manifests so that a poisoned or substituted artifact fails authentication. It supplies exactly the uncontrolled root that Proposition 6 requires and protects the artifact as an object; it does not ask which interventions stay indistinguishable without that root, whether an aggregate or an item-level reference suffices for a given integrity notion, whether a passing substitution changes the conclusion, how to calibrate the decision when only a statistical baseline exists, or what an adaptive attacker can hide from a calibrated sensor. Those questions are treated here; a VAMP-style manifest is one way to instantiate the trusted references this article assumes.

2.3 Calibration of Multi-Sensor Decisions

The family rule of Section III-D is a conformal p-value [20], [21] built on the maximum of per-sensor rank scores, i.e. on Tippett's minimum-p combination [22] with the max-statistic calibration of Westfall and Young [23]; its finite-sample level under exchangeability follows the standard argument, which holds with ties when ties count against rejection [20], [24]. Conformal anomaly detection applies the construction to novelty scores [25], and Bates *et al.* analyse conformal p-values that share one calibration set [26]. We claim no novelty for the rule: its role is to give the decision of an information regime a false-alarm level that holds under a stated premise and to make the executed design's violation of that premise measurable; the asymmetric variant used by an earlier version of this evidence had no such guarantee (Proposition 5(c)).

2.4 Hybrid and Quantum Software Integrity

Volya *et al.* decompose classical-quantum systems into control, compilation, execution and measurement surfaces [1]; surveys extend the threat space to malicious compilation, hardware faults, cloud exposure and side channels [2], [27]; Quantum Vulnerability Factor scores circuit-output sensitivity to faults [28] and Quantum Leak demonstrates timing leakage in cloud services [29]. QCIVET combines

TABLE 1
Positioning against the closest lines of work on nine axes. A dash means that the axis is not the work's declared object, not a deficiency; the hybrid threat taxonomies [1], [2] are descriptive and occupy none of the axes. The last four columns are the axes on which QCIVET, QML-PipeGuard and VAMP exceed this work.

Line of work	Info.-relative blind regions	Eval.-label boundary	Aggregate vs. item reference	Statistical guarantee	Adaptive attacker	Real QPU	Provider / authentication	Drift handling	Deployed / runtime
Stealthy attacks on CPS [3]–[7]	linear dynamics	–	–	residual, impact bounds	model-aware by construction	–	–	–	testbed [7]
Runtime assurance; integrity monitoring [9]–[11]	–	–	object digests	deterministic	–	–	–	–	deployed systems
VAMP [19]	–	signed evaluation sets	object manifests	cryptographic	–	–	✓ signatures	–	–
QCIVET [30]	–	–	reference observables	calibrated deviation tests	–	✓	–	calibrated noise	verification engine
QML-PipeGuard [31]	–	–	channel fingerprint	calibrated tolerance	channel substitution	✓ IBM Heron	–	✓ calibration drift	hardware runs
Evidence sufficiency [15]; eval. blindness [18]	proxy coverage; taxonomy	delayed, silent outcomes	–	–	–	–	–	concept drift vs. proxies	incident corpus
This work	**views; structural vs. calibrated**	**first-class**	**aggregate: conclusion; item: identity**	**conformal; exact under exchangeability**	**executed (F5)**	– simulator	– unsigned hash chain	measured, not handled	– offline, frozen outputs

stage contracts, hash-chained traces, behavioural subtyping and calibrated observable-deviation tests with QPU validation [30]; QML-PipeGuard fingerprints the quantum stage through expectation values under a structured measurement family, absorbs benign calibration drift within a calibrated tolerance and detects channel substitution on an IBM Heron processor [31]; a multi-level framework evaluates circuit-integrity metrics [32]; and a cross-provider analysis proposes a unified provenance contract for quantum jobs [33]. Quantum software testing addresses the oracle problem through differential, metamorphic and equivalence-modulo-input relations [34]–[36], and compiler verification, program logics and runtime assertions supply formal and runtime mechanisms [37]–[41]. We make no priority claim for contracts, hash chains, stage decomposition, semantic probes or fail-closed verification: their unit is the quantum stage or the pipeline trace; ours is the information set that makes a changed asset distinguishable, with the evaluation-label path as a first-class boundary.

Table 1 states the combination claim and its limits: QCIVET, QML-PipeGuard and VAMP carry hardware, drift or authentication evidence that this work does not; this work derives blind regions from views of a learning workflow whose protected boundaries include the evaluation labels, separates a statistical baseline from an authenticated same-batch anchor, and composes the coverage into a decision whose false-alarm level, unsafe-allow count and cost on near-null variation are measured.

2.5 Relation to Companion Work

Three companion studies by the authors are disclosed to delimit the claim. *Sharp target-domain certificates* [42] bounds the advantage of a fixed quantum-kernel candidate under distribution shift (shared data sources and fidelity kernels; an identification interval, not the observability of interventions). *Conditional validity of quantum event classifiers* [43], the closest relative, conditions fail-closed verdicts on a declared information set and exhibits exact sensor blindness to weight-only nuisances in collider data; here the conditioning is applied to the integrity of the evidence chain under interventions, with trusted references, calibrated policy and executable response. *Candidate comparability before promotion* [44] decides whether a challenger should replace a deployed detector after a drift alarm (shared benchmarks and label-free monitors). None of the propositions, interventions, tables or experiments reported here appears in those works.

3 Observation Model and Blind Regions

The complete statement with proofs is in the supplement. The results are elementary by design: their role is to make exact which evidence separates which intervention class, so that the gates test statements rather than intuitions.

3.1 States, Interventions and Materiality

A workflow state separates stored artifacts from those the workflow derives. The primitive artifacts of one evaluation run are

$$s = (X, P, C, E, \xi, g, f, y) \in \mathcal{S}, \tag{1}$$

acquired features X, preprocessing P, circuit or feature map C, execution context E, estimation randomness ξ, post-processing map g, fitted predictor f and evaluation labels y. The derived artifacts are recomputed from them: the representation $\tilde{X} = P(X)$; the semantic kernel $K_{\text{sem}} = \Phi(C)$ induced by the circuit on a fixed probe set; the finite-shot estimate $\hat{K} = \text{est}(K_{\text{sem}}, E, \xi)$; the observed kernel $K_{\text{obs}} = g(\hat{K})$ delivered downstream after estimation, transmission and post-processing; the predictions $\hat{y} = f(\tilde{X}, K_{\text{obs}})$, indexed by item identity $i = 1, \ldots, n$; and the result $R = r(\hat{y}, y)$, balanced accuracy of $\hat{y}$ against y. An intervention a overwrites a declared set of artifacts, primitive or derived, keeps every non-descendant fixed and recomputes the descendants through the workflow with ξ retained; the baseline is s_0 and $s_a = a(s_0)$. Label-only T_y overwrites y item-wise, so only R is recomputed (its prior-preserving subclass T_y^π also preserves the class histogram); feature-side mechanisms overwrite $\tilde{X}$ or X and recompute $\hat{y}$ and R; circuit-side interventions overwrite C and recompute K_{sem} and everything below it; estimation variation overwrites ξ or E with K_{sem} fixed; post-processing overwrites g or K_{obs} with K_{sem} and $\hat{K}$ fixed. "Fixes everything else" thus means every non-descendant. The signed conclusion change is

$$\Delta_R(a) = R(s_0) - R(s_a). \tag{2}$$

An intervention is *material at level* τ if $|\Delta_R(a)| \geq \tau$; $\tau \to 0^+$ means any change of the reported conclusion. Earlier versions of this evidence used the positive part $\max(\Delta_R, 0)$; the signed endpoint is used here because an apparent improvement caused by an integrity failure is a conclusion change.

3.2 Views, Refinement and Trusted References

An information set $\mathcal{I}$ is a view map $V_\mathcal{I} : \mathcal{S} \to \mathcal{O}_\mathcal{I}$; the auditor observes $V_\mathcal{I}(s)$ and nothing else. The regimes studied are $\mathcal{I}_X$ (the multiset of rows of $\tilde{X}$), $\mathcal{I}_{XF}$ (the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$), $\mathcal{I}_{Y_m}$ (the class-count vector of y), $\mathcal{I}_{XFY}$ (the multiset of triples $(\tilde{x}_i, \hat{y}_i, y_i)$) and $\mathcal{I}_Q$ (circuit, kernel and execution evidence). $\mathcal{I} \sqsubseteq \mathcal{I}'$ (refinement) iff $V_\mathcal{I} = \pi \circ V_{\mathcal{I}'}$ for some π; hence $\mathcal{I}_X \sqsubseteq \mathcal{I}_{XF} \sqsubseteq \mathcal{I}_{XFY}$ and $\mathcal{I}_{Y_m} \sqsubseteq \mathcal{I}_{XFY}$, while $\mathcal{I}_{Y_m}$ is incomparable with $\mathcal{I}_X$ and $\mathcal{I}_Q$ is a separate branch. The join $\mathcal{I} \vee \mathcal{W}$ is the pair of views.

A *reference* is stored information about the baseline against which the audited batch is compared; it is *trusted* if no intervention in the class under study can alter it. Level A is a *statistical or historical reference*: a null distribution, a training or reference population, a calibration sample or a baseline score distribution, i.e. information about clean batches in general with no authenticated value of the same batch; every batch-level sensor here holds one (the clean calibration draws and the training-fitted preprocessing). Level B is a *trusted aggregate same-batch reference*, a permutation-invariant summary $\rho = V_\mathcal{J}(s_0)$ of the audited batch such as $M(s_0)$, $\text{hist}(y_0)$ or $R(s_0)$. Level C is a *trusted item-aligned same-batch reference* such as $(y_{0,i})_i$. The distinction that carries the results is between a statistical baseline (A) and an authenticated same-batch anchor (B, C), not between having and lacking a reference. For a label-path intervention, *item-identity integrity* holds if $y_a = y_0$ item-wise, *aggregate integrity* if $M(s_a) = M(s_0)$ and *conclusion integrity* if $R(s_a) = R(s_0)$; each implies the next and no converse holds. $\mathcal{I}^\star$ denotes a regime augmented with same-batch references of its own baseline view ($\mathcal{I}^\star_{XFY}$ holds the aggregate confusion profile and the item-aligned labels). A label available in $\mathcal{I}_{XFY}$ is not thereby trusted: if the label store is the asset under attack, the auditor sees y_a, not y_0.

3.3 Equivalence, Sensors and Blind Regions

Definition 1. $s \sim_\mathcal{I} s'$ iff $V_\mathcal{I}(s) = V_\mathcal{I}(s')$. A sensor admissible in $\mathcal{I}$ is a measurable $S : \mathcal{O}_\mathcal{I} \to \mathbb{R}^k$, written $S(s) = S(V_\mathcal{I}(s))$.

Lemma 1 (structural blindness). If $s_a \sim_\mathcal{I} s_0$ then $S(s_a) = S(s_0)$ for every sensor admissible in $\mathcal{I}$ (path-wise for a seeded randomized sensor; in distribution for a fresh seed).

Definition 2. For a class $\mathcal{A}$, the structural blind region of $\mathcal{I}$ is $B_\mathcal{I}(\mathcal{A}) = \{a \in \mathcal{A} : a(s_0) \sim_\mathcal{I} s_0\}$ and the sensor blind region of a family $\mathcal{S}$ is $B_{\mathcal{I},\mathcal{S}}(\mathcal{A}) = \{a : S(a(s_0)) = S(s_0)\ \forall S \in \mathcal{S}\}$. By Lemma 1, $B_\mathcal{I} \subseteq B_{\mathcal{I},\mathcal{S}}$, with equality iff $\mathcal{S}$ is *baseline-separating* on the orbit of $\mathcal{A}$ (a different value at $a(s_0)$ than at s_0 whenever the views differ); detection needs baseline separation, identification would need pairwise separation, which is not claimed. Sensor blindness thus decomposes into information-level blindness and sensor insufficiency; the adaptive attacker of Section VI-D exploits the second. The structural and sensor auditability gaps at level τ are

$$G_\mathcal{I}(\tau) = \{a : |\Delta_R(a)| \geq \tau,\ a \in B_\mathcal{I}\}, \quad G_{\mathcal{I},\mathcal{S}}(\tau) \supseteq G_\mathcal{I}(\tau). \tag{3}$$

Proposition 1 (monotonicity). If $\mathcal{I} \sqsubseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_\mathcal{I}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_\mathcal{I}(\tau)$ for every τ.

Proof: $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ implies $V_\mathcal{I}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_\mathcal{I}(s_0)$. □

Corollary 1 (label-path boundaries). Let f be fixed and deterministic. (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$: every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change. (b) $T_y^\pi \subseteq B_{\mathcal{I}_{Y_m}}$: every sensor admissible in $\mathcal{I}_{Y_m}$ is invariant under a prior-preserving change. These are the two propositions of the earlier version of this evidence, now corollaries.

Proposition 2 (closure). $B_{\mathcal{I} \vee \mathcal{W}}(\mathcal{A}) = B_\mathcal{I}(\mathcal{A}) \cap B_\mathcal{W}(\mathcal{A})$; a class $\mathcal{A} \subseteq B_\mathcal{I}$ becomes completely separable after adding $\mathcal{W}$ iff $V_\mathcal{W}(a(s_0)) \neq V_\mathcal{W}(s_0)$ for every $a \in \mathcal{A}$. Consequently no view factoring through $(\tilde{X}, f(\tilde{X}), \text{hist}(y))$ separates any $a \in T_y^\pi$; the multiset $\mathcal{I}_{XFY}$ separates it unless labels are permuted only among items with identical $(\tilde{x}, \hat{y})$, and $\mathcal{I}^\star_{XFY}$ separates every non-identity relabeling.

Proposition 3 (materiality forces aggregate separability). Let $R = g(M)$ be a function of the confusion matrix $M(s)$, as balanced accuracy is. If $a \in T_y$ and $\Delta_R(a) \neq 0$ then $M(s_a) \neq M(s_0)$; hence a is separable in the confusion view and in $\mathcal{I}_{XFY}$, and $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for every $\tau > 0$.

Proof: Contrapositive of $M(s_a) = M(s_0) \Rightarrow R(s_a) = R(s_0)$; the confusion view is a coarsening of the multiset of triples. □

The proposition holds for every metric that is a function of the binarised confusion matrix; for a ranking metric such as ROC-AUC the same argument holds with the multiset of (score, label) pairs as the aggregate (supplement). No AUC experiment is run.

3.4 Three Auditor Classes and Decision-Level Calibration

A *reference-anchored* auditor holds a trusted same-batch reference ρ (level B or C) and computes $D(s) = d(V_\mathcal{J}(s), \rho)$ with a metric d; then $D(s_0) = 0$ exactly, the rule "fire iff $D > 0$" has zero false-alarm probability, and separability in $\mathcal{J}$ is sufficient for detection. A *batch-level* auditor holds a level-A reference only, a null model of $V_\mathcal{I}$ over fresh clean batches, and decides with a calibrated test at budget α; for a separable a its detection probability is the power against the shift relative to batch-to-batch variability, and separability is not sufficient. A *post-hoc certifier* recomputes a view from inputs it trusts (the evaluation contract recomputes R from $(\hat{y}, y)$) and needs trusted inputs rather than a stored value.

Proposition 4. (i) For any auditor whose decision is a function of $V_\mathcal{I}(s)$, $a \in B_\mathcal{I}$ implies $\text{fire}(s_a) = \text{fire}(s_0)$ on every batch, so the calibrated detection rate equals the false-alarm rate. (ii) A reference-anchored auditor detects every $a \notin B_{\mathcal{J}}$, deterministically.

Corollary 3 (which reference certifies which integrity). Let $a \in T_y$ with f fixed. (a) With a trusted aggregate reference $M_0 = M(s_0)$ of the same batch, "fire iff $M(s_a) \neq M_0$" detects every violation of aggregate integrity and, by Proposition 3, every material a: conclusion and aggregate integrity need only level B. (b) Every reference that factors through M, $\mathrm{hist}(y)$ or R is invariant under relabelings that permute labels among items with equal predictions (C1); certifying item identity needs level C, and an item-aligned y_0 detects every non-identity relabeling.

Proposition 5 (union versus conformal family calibration). (a) If m sensors fire under the null with probabilities $p_j \leq \alpha$, then $\max_j p_j \leq \Pr_0[\bigcup_j E_j] \leq \min(1, \sum_j p_j) \leq \min(1, m\alpha)$, the bounds being attained by nested and disjoint events. (b) Let $v_1, \ldots, v_n$ be the sensor vectors of n calibration draws and v_{n+1} that of the audited batch; give every member j of the augmented set the leave-one-out family score $\tilde{U}_j = \max_s \#\{i \neq j : v_{s,i} < v_{s,j}\}/n$, let $p = (1 + \#\{i \leq n : \tilde{U}_i \geq \tilde{U}_{n+1}\})/(n+1)$, and fire iff $p \leq \alpha$ (ties against firing). At most $\lfloor \alpha(n+1) \rfloor$ members of any augmented set would fire if audited; hence, if the $n+1$ draws are exchangeable, $\Pr_0[p \leq \alpha] \leq \lfloor \alpha(n+1) \rfloor/(n+1) \leq \alpha$ with no continuity or tie assumption [20], [22], [23]; $10/201 = 0.0498$ for $n = 200$, $\alpha = 0.05$. For one sensor without ties the rule is the per-sensor rule of Gate N. The level is marginal over the joint draw of calibration set and audited batch; nothing is claimed when the premise fails. (c) The asymmetric rule of the earlier version, which scored calibration draws against the other $n-1$ draws only, has no such guarantee: with three sensors and cyclic orderings over five vectors it fires with probability 0.60 at $\alpha = 0.2$ under exchangeability (supplement), and on the frozen draws it exceeds α under exchangeable re-splits where the conformal rule does not (Section VI-B).

Proposition 6 (no local authentication without an uncontrolled root). Let a local verifier be any map $\mathrm{acc}(V_{\mathcal{I}}(s), \rho)$ of the observed view and a reference bundle, with honest references $\rho_H(s) = V_{\mathcal{J}}(s)$. If it accepts every honest pair and the class can produce $(V_{\mathcal{I}}(s'), \rho_H(s'))$ for some honest $s' \not\sim_{\mathcal{I}} s_0$ (joint control), the substitution is accepted as an honest run of s': no function of that input establishes authenticity relative to s_0, and if $R(s') \neq R(s_0)$ a material change is served. A bundle component $V_{\mathcal{K}}(s_0)$ the class cannot write rejects every substitution with $V_{\mathcal{K}}(s') \neq V_{\mathcal{K}}(s_0)$ (Proposition 4(ii)): authenticity requires a root outside the declared class. The result concerns authenticity, not consistency; by Proposition 3 a material substitution does change the joint view.

3.5 Counterexamples

Eight minimal constructions separate the events a robustness number conflates; the supplement tabulates them with the frozen expansion observations that realise each. C1 swaps the labels of two items with equal predictions: M, the histogram and R are invariant while item identity changes (808 of 3,600 label rows), an integrity violation with no conclusion impact, visible only item-wise against a trusted reference. C2–C4 separate marginal invariance, confusion-matrix invariance and conclusion invariance, C5–C6 do the same on the feature side, C7 shows that the clipped "harm" endpoint of the earlier evidence hid 1,534 conclusion changes (433 on the label path), and C8 is the empirical face of Proposition 3 (2,617 of 2,617 material label rows change M). Of the 3,600 label rows, a trusted aggregate reference of the same batch detects 2,792 (every material one, Corollary 3a) and the remaining 808 are C1 relabelings that only an item-aligned reference exposes (Corollary 3b). A ninth construction, the cluster-preserving drift of Section VI-D, is not a structural blind region but sensor insufficiency against an adaptive attacker, measured rather than proved.

3.6 The Quantum Branch as an Instance of the Lattice

Let $\mathcal{C}$ be the circuit representations (canonical OpenQASM 3 with bound parameters), $\Phi : \mathcal{C} \to \mathcal{K}$ the semantic map to the ideal fidelity kernel on a fixed probe set, $K_{\mathrm{sem}} = \Phi(C)$, $\hat{K}$ and K_{obs} the estimated and observed kernels of Section III-A, $h(C)$ the provenance hash, $A(\cdot)$ the algebraic invariants and $f_K(\tilde{X})$ the decision computed from a kernel. Three intervention classes are kept apart: (a) circuit-side, overwriting C; (b) estimation variation, with C and K_{sem} fixed and $\hat{K}$ changing; (c) post-processing or kernel substitution, with C, K_{sem} and $\hat{K}$ fixed and K_{obs} changing. Each inclusion below names its class; none is claimed across classes.

Proposition 7 (quantum lattice). Assume h collision-free on the circuits under study. (i) [class (a)] $[C] = \Phi^{-1}(\Phi(C))$ is the semantic class of C; approved transpilation and common-unitary rewrites map C into $[C]$ without fixing $h(C)$, so provenance refines the semantic view, $B_h \subseteq B_{K_{\mathrm{sem}}}$, strictly whenever the class contains an approved rewrite: separability in provenance is not harm, and the auditor needs the approved class, implemented as semantic equality of probe kernels against the trusted $K_{\mathrm{sem},0} = \Phi(C_0)$ (level B). (ii) [class (c)] A and f_K coarsen the observed kernel, so $B_{K_{\mathrm{obs}}} \subseteq B_A$ and $B_{K_{\mathrm{obs}}} \subseteq B_f$; a PSD-preserving substitution K' with $A(K') = A(K_{\mathrm{obs},0})$ lies in $B_A \setminus B_{K_{\mathrm{obs}}}$ and, since $h(C)$ and the semantic probe of the unchanged circuit see nothing, is closed only by an anchored comparison of K_{obs} itself: the trusted semantic probe and the trusted observed-kernel reference are different anchors. A kernel change that crosses no decision boundary lies in $B_f \setminus B_{K_{\mathrm{obs}}}$. (iii) [class (b)] For an honest finite-shot estimate, exact equality $\hat{K} = K_{\mathrm{sem},0}$ has false-alarm probability $1 - \Pr[\hat{K} = K_{\mathrm{sem},0} \mid \text{honest}]$, which depends on the discrete support of the estimator and may be large or equal to one but is not universally one (fidelity $1/2$ at two shots gives $1/2$). Exact equality is therefore not an appropriate acceptance criterion; $d(\hat{K}, K_{\mathrm{sem},0})$ must be treated as a level-A statistic with a null from repeated honest estimation, to which Proposition 5(b) applies with one sensor when the repeated and audited estimates are exchangeable, and where no such null is calibrated no statistical integrity claim is made. The proposition adds no quantum mechanics: it names the two features the classical branches lack, an approved equivalence class coarser than provenance and an estimation step that turns an exact anchor into a statistical one. Hash chaining of the audit envelope is a reference-anchored comparison over the record itself; its root is not assumed uncontrolled, so Proposition 6 applies.

TABLE 2
Adversary and failure model of the executed interventions. Roots: assets the class cannot write. Identifying regime: the least evidence that separates the class in the frozen design under the conformal family rule; "exact" means an item-aligned trusted reference; (a)–(c) are the intervention classes of Proposition 7.

Mechanism and capability (boundary)	Class	Roots it cannot write	Identifying regime	Supported claim
Random label flip: rewrite r of y at random (evaluation/report)	Integrity corruption	X, P, C, K, f; item-aligned $y^{\star}$ if declared	exact only; $\mathcal{I}_{Y_m}/\mathcal{I}_{XFY}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}$ (Cor. 1a); undetected at budget by marginal or batch joint evidence at these rates
Prior-preserving label flip: pairwise swaps preserving class counts (evaluation/report)	Adaptive attacker	as above	exact only; $\mathcal{I}_{XFY}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}, \mathcal{I}_{Y_m}$ (Cor. 1); every material instance changes M (Prop. 3)
Confusion-preserving relabeling: swaps among equal predictions (realised, not executed)	Adaptive attacker	as above	item-aligned reference only	item-identity violation without aggregate or conclusion impact (C1; 808 rows)
Feature sign flip: negate p of $\tilde{X}$ entries (acquisition/preprocessing)	Integrity corruption	training data, fitted model, reference features	$\mathcal{I}_X$ 0.09–0.72; $\mathcal{I}_{XF}$ 0.13–0.84; exact 0.79–0.91	partially separable; strength-dependent residual
Mean shift, scaling drift: shift or scale all features (acquisition/preprocessing)	Fault robustness	as above	$\mathcal{I}_X$ 0.96–1.00 by cluster fingerprint	detected in this design by a fragile fingerprint
Feature dropout: impute p of entries (preprocessing)	Fault robustness	as above	batch 0.00–0.03; exact 0.49–0.76	separable in $\mathcal{I}_{XF}$ yet statistically undetectable with a level-A reference
Cluster-preserving mean shift and scaling: same mechanisms on the 32% of entries outside tight clusters (executed, Gate A)	Adaptive attacker	as above	$\mathcal{I}_X$ 0.01–0.56, $\mathcal{I}_{XF}$ 0.06–0.66 at matched strengths; exact for every material row	evades the fingerprint while keeping 83–91% of the conclusion changes; served by P2 in 28–39% of material rows
Circuit-parameter mutation; repetition change: alter C before execution (design; (a))	Adversarial attack	reference circuit hash, probe kernel $K_{\mathrm{sem},0}$	circuit hash and semantic probe 15/15	identified by anchored provenance and semantics; outputs miss subdecision changes
Benign transpilation; common-unitary rewrite: compiler pass ((a))	Approved equivalence	–	hash changes 15/15, K_{sem} unchanged	separability in $h(C)$ is not harm (Prop. 7(i))
Asymmetric, diagonal-eroded kernel edit: edit K_{obs} after estimation ((c))	Integrity corruption	reference kernel, algebraic invariants	algebra 15/15	detected by structural invariants of K_{obs}
PSD-preserving kernel substitution: valid-looking K_{obs} ((c))	Adaptive attacker	reference kernel	algebra 0/15; circuit hash 0/15; anchored K_{obs} comparison 15/15	sensor-level blindness closed only by a reference on the observed kernel (Prop. 7(ii))
Binomial shot emulation: finite-shot estimate (execution; (b))	Fault robustness	–	repeated estimation 15/15 in these cells	held for adjudication; exact equality is not the criterion (Prop. 7(iii)); not hardware evidence
Post-hoc envelope edit: rewrite stored contracts (record)	Adversarial attack	none assumed	hash chain	tamper-evident only; no identity or non-repudiation

4 Adversary and Failure Model

Table 2 assigns every executed intervention to one of four classes with the roots it cannot write and the regime that identifies it (complete per-class record in the artifact). Evaluation-label classes stand for corruption of the ground-truth store or label join; feature-side classes for corruption or drift of the acquisition path feeding both branches; the cluster-preserving class for an attacker who knows the batch-level fingerprint; circuit and kernel classes for corruption of the circuit/parameter store, the compiler output or the post-estimation kernel; shot emulation for legitimate estimation uncertainty. Equal weighting is a stress profile, not a threat distribution; no prevalence is estimated.

The evidence trusts the local runtime and host, the training data and fitted model, the reference input and preprocessing declaration, the reference circuit and probe kernels, SHA-256 collision resistance, the item-level label reference only where $\mathcal{I}^{\star}_{XFY}$ is declared, and the code that builds and verifies the envelope. Compromise of the operating system, runtime or verifier; provider identity, forged job or calibration metadata and provider attestation; malicious schedulers, physical mapping and unapproved passes on hardware; device-calibrated noise, crosstalk, multi-tenant interference and QPU faults; side channels and circuit confidentiality; denial of service; and the operational Fleet Management System are outside this study.

5 Methodology

5.1 Data, Pipelines and Interventions

The frozen first gate uses a balanced binary subset of CICIDS2017 [45]: 3,000 rows, 77 numeric columns and 128/128 capped training/evaluation samples. The expansion adds eight fixed environments E1–E8 (supplement): five from CICIDS2017 (a 256/256 scale gate and four temporal source-target pairs), two from UNSW-NB15 [46] (a balanced ID subset and a temporal pair) and one from ToN-IoT [47] (a balanced ID subset), all at 128/128 unless stated and projected to dimensions 8, 10 and 12. CICIDS2017 therefore carries six of the nine environments including Gate 1; the environments are not a probability sample of deployments. Train-defined preprocessing is applied without evaluation labels: TruncatedSVD per branch, training-fitted standardization for the RBF support-vector classifier (SVC) and frozen angular scaling to $[0, 2\pi]$ for the fidelity-kernel branches (ZZ, PauliXYZ and a Z/PauliXZ-equivalent profile, one repetition). Gate 1 uses the Qiskit Machine Learning reference evaluator [48]; the expansion uses a tagged exact-statevector evaluator validated against it at 10^{-10} on unit matrices and by an end-to-end replay (supplement); neither is shot-noise or QPU evidence. The intervention suite has six mechanisms at strengths 0.02, 0.05 and 0.10: feature sign flip, feature-wise mean shift, scaling drift, feature dropout with median imputation, random evaluation-label flip and prior-preserving evaluation-label flip. Feature-only sensors are feature Jensen–Shannon divergence (JSD), maximum mean discrepancy (MMD) and the Kolmogorov–Smirnov (KS) rejection rate against the clean reference batch of the same items; prediction-aware evidence adds score JSD, prediction-rate shift and prediction JSD; label-marginal evidence uses prior shift and label JSD; joint-outcome evidence uses two confusion-profile statistics; reference-anchored sensors (confusion-profile deltas against a trusted aggregate of the same batch; prediction disagreement and item-level label mismatch against item-aligned references) exist only with a trusted reference of the same items and have an exact-zero null.

5.2 Experimental Units and Uncertainty

Gate 1 crosses five split seeds with four nested model seeds; its 7,980 raw rows contain 3,420 exact repeated SVC rows, leaving 4,560 unique observations. The expansion crosses the same five split seeds with two model seeds; all 360 job pairs completed and 13,680 raw rows contain 2,280 verified repeats, leaving 11,400 unique observations. Coverage endpoints are exact counts in the prespecified design; for the secondary model-profile endpoint (supplement) the five split seeds are the inferential units with two-sided 95% Student-t intervals (four degrees of freedom), describing within-environment resampling only.

5.3 Null Calibration and Decision-Level Calibration

Gate N (preregistered in artifact 1.1.0) calibrates the ten distributional sensors per (environment, dimension, model) cell at a nominal per-sensor $\alpha = 0.05$: the threshold is the 191st of 200 clean calibration draws from one half of the held-out evaluation pool, and false alarms are measured on 200 draws from the disjoint other half. Gate F calibrates each regime as a family with the conformal rule of Proposition 5(b) at $\alpha = 0.05$ per decision on the calibration draws only; the earlier asymmetric rule (Proposition 5(c)) and the union rule are computed on the same draws for comparison, the pooled excess of the earlier rule over the nominal level is decomposed into rule bias (its rate minus the conformal rate on identical draws) and design effect (the conformal rate minus nominal), and both rules are also evaluated under 30 exchangeable random re-splits of the 400 pooled draws of every cell; the protocol, its amendment and the expectations recorded before regeneration are in the artifact. Because the 20 draws of a run are overlapping subsets of one pool half, the inferential unit for false-alarm rates is the (environment, split-seed) cluster: pooled rates are descriptive and intervals are five-cluster t intervals per environment. E1 (256-row draws from 322-row halves) is kept in the primary aggregate as preregistered and also reported separately.

5.4 Decision Layer and Offline End-to-End Evaluation

Gate D composes sensors, calibrated rules, the information regime, the trusted-reference status, a materiality threshold and the declared residual blind region into one `allow/hold/block` decision. `block` is reserved for violations of an exact invariant against a trusted reference; `hold` answers statistical evidence or, under P3, missing coverage; the decision composes with the four frozen contracts by the maximum in `allow` < `hold` < `block`. Four policies are fixed and named by what they are: P0 serve-always, the baseline; P1, the uncalibrated union rule of Gate N (risk-tolerant); P2, the conformal family rule, a calibrated *risk-tolerant* policy that may serve under an explicitly declared residual blind region; P3, the *coverage-complete abstaining* variant, which holds whenever a mandatory protected boundary (feature, prediction, label) has neither exact nor declared statistical coverage under the regime. P3 fails closed on missing coverage, not on insufficient power: a boundary covered by a low-power calibrated sensor is served as under P2, so P3 guarantees no minimum detection power. The prototype's contracts fail closed on their invariants; P2 does not and is never called fail-closed. Five regimes are evaluated: $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$, $\mathcal{I}_{XFY}$ (batch-level, level-A references) and $\mathcal{I}^{\star}_{XFY}$ (trusted aggregate and item-aligned references of the same batch). The observation set is frozen: 12,000 clean evaluation draws, 1,200 near-null synthetic controls applied in place (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$; not operational benign traffic), and the 10,800 intervened observations of the eight environments, 7,008 of which change the reported balanced accuracy (primary materiality; $\tau = 0.02$ and 0.05 as sensitivity). The primary endpoint is the unsafe allow: a material observation decided `allow`. The cost endpoints are the clean false-action rate (clean draws not allowed; for $\mathcal{I}^{\star}_{XFY}$ the denominator is the 1,200 exact-zero rows) and the interruption rate on the near-null controls (the same 1,200 rows for every regime), decomposed into statistical holds and exact-reference blocks. The evaluation is offline, on frozen outputs, not a deployed runtime service; twelve consistency checks fail closed, and every table and figure is generated from the manifested outputs.

5.5 Adversarial Gate: Cluster-Preserving Perturbation

Gate A (preregistered before execution) executes the adaptive attacker that the earlier version only discussed. The standardized projected features hold tight clusters of rows (in every environment some feature keeps 69–87% of the rows within 0.01 standard deviations) that a fresh clean batch reproduces and any in-place displacement smears, which is why mean shift and scaling drift were detected in every cell. The attacker holds the batch, knows the sensor definitions and the fingerprint, and applies the executed mechanisms unchanged to every unclustered entry (an entry is clustered when at least $\lceil 0.05n \rceil$ rows, never fewer than two, lie within 0.01 batch standard deviations of it); clustered entries and labels are untouched. Strengths 0.02, 0.05, 0.10 match the frozen suite and 0.25, 0.50 probe whether materiality can be bought on the unclustered entries; the executed mechanisms run in the same jobs as matched controls and must reproduce the frozen expansion exactly. The gate runs in the eight environments with the frozen maps, caps and seeds (240 exact-statevector jobs) and is scored with the calibration draws, rules and policy layer of Gates F and D; window, minimum mass and strength grid were fixed before execution.

5.6 Quantum Gate and Executable Prototype

A separate simulator gate isolates the quantum boundary with CICIDS, 64/64 samples, five split seeds, dimensions 4, 6 and 8 and 11 conditions (165 cells): clean, benign level-1 transpilation and a common post-feature-map rewrite as controls; data-dependent RZ mutations at 0.02 and 0.10; a feature-map repetition change; asymmetric, diagonal-eroded and PSD-preserving kernel edits; and binomial fidelity-estimation emulators at 256 and 1,024 shots. Each cell records canonical OpenQASM 3 provenance, kernel hashes, the anchored comparison against the trusted reference kernel, algebraic checks, repeated-estimation discrepancy and prediction changes. The prototype instantiates four ordered contracts (input/preprocessing, circuit/kernel, execution/result, evaluation/report) in an SHA-256 chain over one compact CICIDS cell and six prespecified scenarios; it evaluates composition and decision logic, not throughput.

6 Results

6.1 Exact Label-Path Blind Regions

The zero-response cells are validation checks of the structural prediction, not findings: Corollary 1 says that a sensor whose input does not change cannot respond, and the checks confirm that the implementation realises the declared views. All 3,600 expansion and 1,440 Gate-1 evaluation-label observations (of 11,400 and 4,560 unique observations) have zero prediction disagreement and exactly unchanged feature JSD, MMD, KS rejection and score JSD; all 1,800 and 720 prior-preserving observations also have zero prior shift and label JSD; the verifier recomputes every count. The informative quantity is how often these blind regions coincide with material conclusion changes and, in Section VI-C, with serving decisions. The signed conclusion change is not one-sided: in the expansion 2,184 label observations lower the reported balanced accuracy, 983 leave it unchanged and 433 raise it (Gate 1: 1,276, 105, 59). Every one of the 2,617 (Gate 1: 1,335) label observations with a changed conclusion has non-zero item-aligned confusion evidence, and every one of the 2,184 with a lowered conclusion, the count reported by the earlier version, is among them.

6.2 Decision-Level Calibration

Over the 12,000 evaluation draws the per-sensor false-alarm rates lie between 0.029 and 0.069, but the batch-level rule "fire if any sensor fires" is not calibrated at the level of the decision: its rates are 0.125, 0.203, 0.048 and 0.259 for $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$ and $\mathcal{I}_{XFY}$ (three, six, two perfectly dependent and ten sensors). The conformal family rule brings them to 0.056, 0.058, 0.048 and 0.053 (Fig. 1(a)), observed rates in a design that violates the exchangeability premise. The asymmetric rule of the earlier version gave 0.061, 0.073, 0.048 and 0.079 on the same draws and attributed its whole excess over 0.05 to the design. That attribution was wrong: the rule bias is +0.005, +0.015, +0.000 and +0.025, the design effect +0.006, +0.008, -0.003 and +0.003, and under exchangeable re-splits of the pooled draws, where any excess is the rule's own, the earlier rule fires at 0.052, 0.058, 0.040 and 0.059 while the conformal rule stays at 0.044, 0.042, 0.040 and 0.036 against its level 0.0498. What remains with the conformal rule is the design effect, concentrated in E1, whose 200 draws per cell carry the information of about one independent batch per pool half (0.125, 0.115, 0.123 in $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{XFY}$); without E1 the conformal rule sits at 0.048, 0.052, 0.048 and 0.045; E1 stays in the primary aggregate. A split construction (0.049, 0.039, 0.067, 0.034; supplement) is not adopted.

As a consistency check of Proposition 4(i), the calibrated rules confirm the structural prediction: on all 3,600 label observations the per-sensor and the conformal $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules, and on all 1,800 prior-preserving observations the $\mathcal{I}_{Y_m}$ rules, fire zero times, as they must when their input is unchanged. The informative result is on the other side: the batch-level $\mathcal{I}_{XFY}$ rules detect 43 (union) and 11 (conformal) of the 2,617 material label observations, 0.016 and 0.004, although every one of them is separable (Proposition 3, C8): a 2–10% relabeling of a 128-row batch shifts the confusion profile by less than its variability across fresh clean batches. A trusted aggregate reference of the same batch detects all 2,617 exactly (Corollary 3a) and 2,792 of the 3,600 label rows in total; the 808 relabelings that preserve every aggregate (C1) are exposed only by the item-aligned label reference (Corollary 3b). Feature-side coverage is mechanism-dependent: mean shift and scaling drift are detected in 0.96–1.00 of cells because the standardized projected features hold tight clusters that an in-place perturbation smears and a fresh batch reproduces, a legitimate but fragile fingerprint that also fires on the near-null controls (40–64% with the conformal rule); feature dropout with imputation is nearly invisible at the batch level (0–3%) yet changes predictions in 49–76% of cells. Calibration costs little detection: containment in $\mathcal{I}_{XF}$ is 0.40 under the union and 0.38 under the conformal rule at the primary threshold, 0.70 versus 0.66 at $\tau = 0.05$.

6.3 Offline End-to-End Decisions

Table 3 and Fig. 1(b,c) report the offline end-to-end evaluation. Under P0 every one of the 7,008 material observations is served. Under the batch-level regimes the conformal policy P2 serves 4,496 ($\mathcal{I}_X$), 4,365 ($\mathcal{I}_{XF}$) and 4,390 ($\mathcal{I}_{XFY}$) of them with clean false-action rates 0.056, 0.058 and 0.053, whereas the uncalibrated union serves 4,429, 4,231 and 4,180 at 0.125, 0.203 and 0.259: the union buys 0.40 versus 0.37 containment in $\mathcal{I}_{XFY}$ at more than four times the clean false-action rate and 0.76 versus 0.49 interruption of the near-null controls. The earlier asymmetric rule served 4,494, 4,327 and 4,322; the correction moves the count by less than 1% of the material rows. The unsafe allows are dominated by structure, not by thresholds: all 2,617 material label observations are served under $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ (residual blind region), 2,606 of them under batch-level $\mathcal{I}_{XFY}$ (statistically undetectable), and 1,057 of 1,079 material dropout observations under $\mathcal{I}_{XF}$; $\mathcal{I}_{Y_m}$ contains nothing in this suite.

The trusted-reference regime serves none of the 7,008 material observations with zero clean false actions on its 1,200 exact-zero rows (the aggregate reference alone would suffice, by Corollary 3a), and blocks every non-identity relabeling and every prediction change through its item-

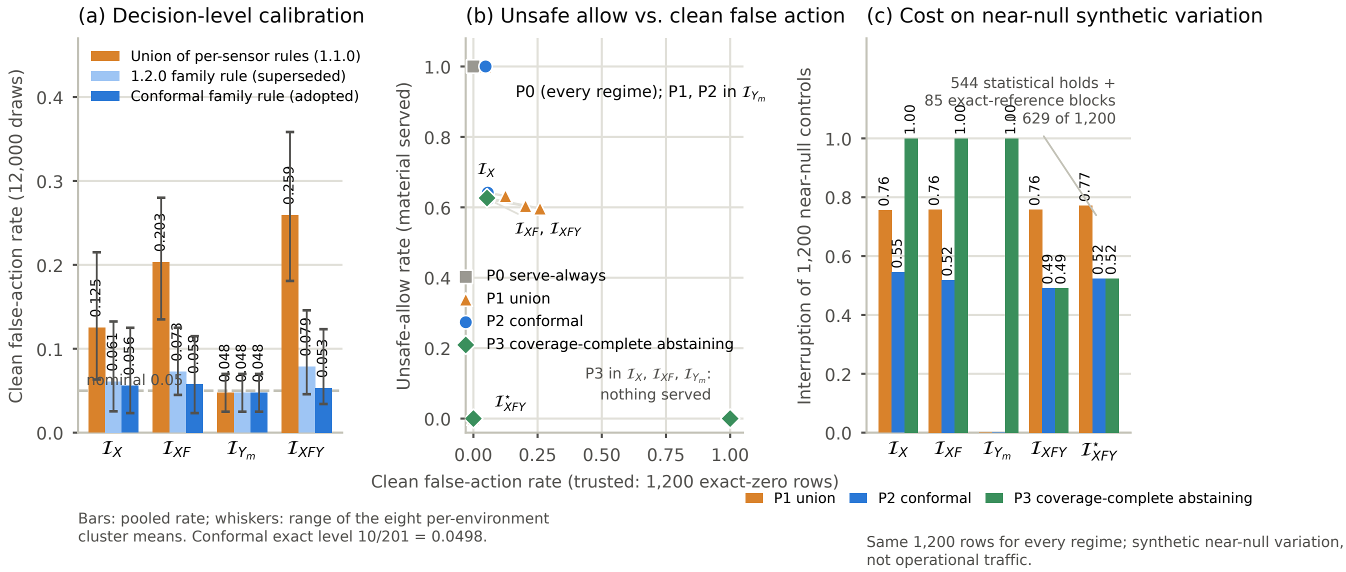

Fig. 1. Gates F and D (frozen evidence of artifact 1.3.0). (a) Decision-level false-alarm rate on 12,000 disjoint clean draws of the union rule, of the superseded asymmetric rule and of the conformal family rule per regime; whiskers span the eight per-environment cluster means. (b) Unsafe-allow rate against clean false-action rate for every policy and regime at the primary materiality threshold; the trusted regime is the only point at the origin, its clean denominator is the 1,200 exact-zero rows, and the coverage-complete abstaining policy P3 serves nothing where the label boundary has no coverage. (c) Interruption of the same 1,200 near-null synthetic controls by each policy and regime; the trusted regime's origin in (b) costs 629 controls here, 544 statistical holds and 85 exact-reference blocks.

aligned references. Its cost has two parts. Of the 1,200 near-null synthetic controls the trusted calibrated policy interrupts 629: 544 are statistical holds by its batch-level component, the specificity cost of the cluster fingerprint that P2 already pays in the batch regimes (590–654 holds), and 85 are exact-reference blocks of controls whose predictions changed under a 0.001 perturbation. Exact provenance therefore adds 85 of 1,200 to a statistical cost the calibrated batch-level policy incurs anyway; the total is 0.52 against 0.52 for P2 in $\mathcal{I}_{XF}$ (Fig. 1(c)). The controls are synthetic near-null variation, not operational traffic, so neither number is a production false-positive rate, and the two denominators of Table 3 are reported side by side. The coverage-complete abstaining policy P3 makes information-set conditionality literal: in $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and $\mathcal{I}_{Y_m}$ the label boundary has no coverage, so nothing is served and every clean and near-null row is interrupted; in $\mathcal{I}_{XFY}$ and $\mathcal{I}^{\star}_{XFY}$ every boundary is covered and P3 coincides with P2, including P2's 4,390 unsafe allows in $\mathcal{I}_{XFY}$: abstention on missing coverage is not a guarantee of power, and the choice between P2 and P3 is a declared risk decision, not a detector property. The composition with the six frozen contract envelopes reproduces their actions (supplement).

TABLE 3
Offline end-to-end decisions on the frozen observations (primary materiality $|\Delta_R| > 0$; 7,008 material of 10,800 intervened observations). P1 union, P2 conformal (calibrated, risk-tolerant), P3 abst. (coverage-complete abstaining). FPR: clean false-action rate on 12,000 disjoint clean draws, or on the 1,200 exact-zero rows for $\mathcal{I}^{\star}_{XFY}$ (a different denominator). Benign: fraction of the 1,200 near-null synthetic controls held or blocked (same rows for every regime). P0 serve-always allows all 7,008 material observations in every regime. Blind: material observations structurally indistinguishable from the reference under the regime.

Regime	Policy	FPR	Benign	Unsafe	Contain.	Blind
$\mathcal{I}_X$	P1 union	0.125	0.76	4,429	0.37	2,617
$\mathcal{I}_X$	P2 conf.	0.056	0.55	4,496	0.36	2,617
$\mathcal{I}_X$	P3 abst.	1.000	1.00	0	1.00	2,617
$\mathcal{I}_{XF}$	P1 union	0.203	0.76	4,231	0.40	2,617
$\mathcal{I}_{XF}$	P2 conf.	0.058	0.52	4,365	0.38	2,617
$\mathcal{I}_{XF}$	P3 abst.	1.000	1.00	0	1.00	2,617
$\mathcal{I}_{Y_m}$	P1 union	0.048	0.00	7,008	0.00	5,450
$\mathcal{I}_{Y_m}$	P2 conf.	0.048	0.00	7,008	0.00	5,450
$\mathcal{I}_{Y_m}$	P3 abst.	1.000	1.00	0	1.00	5,450
$\mathcal{I}_{XFY}$	P1 union	0.259	0.76	4,180	0.40	0
$\mathcal{I}_{XFY}$	P2 conf.	0.053	0.49	4,390	0.37	0
$\mathcal{I}^{\star}_{XFY}$	P2/P3 (exact)	0.000	0.52	0	1.00	0

6.4 Adaptive Cluster-Preserving Attacker

Gate A ran as preregistered: 240 of 240 jobs, and the clean rows and matched controls reproduce the frozen expansion rows exactly. The attacker perturbs 0.32 of the entries on average (0.12–0.55 across environments, branches and dimensions); Fig. 2 and the supplement report the outcome. At the matched strengths 0.02, 0.05 and 0.10 the conformal rule detects the executed mean shift in 1.00, 1.00 and 1.00 of the rows in $\mathcal{I}_X$ and its cluster-preserving variant in 0.01, 0.08 and 0.34 (0.06, 0.24 and 0.48 in $\mathcal{I}_{XF}$); scaling drift behaves alike (0.05–0.56 in $\mathcal{I}_X$ against 0.99–1.00). The attacker keeps most of the harm: the material fraction of the cluster-preserving mean shift is 0.29, 0.40 and 0.54 against 0.33, 0.46 and 0.59. Detection recovers at 0.25 and 0.50, where the unclustered entries move far enough to shift the feature distribution against the between-batch null. The classical branch (standardized features) is more evasive than the quantum branch (features scaled to $[0, 2\pi]$): at strength 0.10 the mean shift is detected in 0.30 and 0.34 of classical cells and 0.37 and 0.58 of quantum cells in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$, with

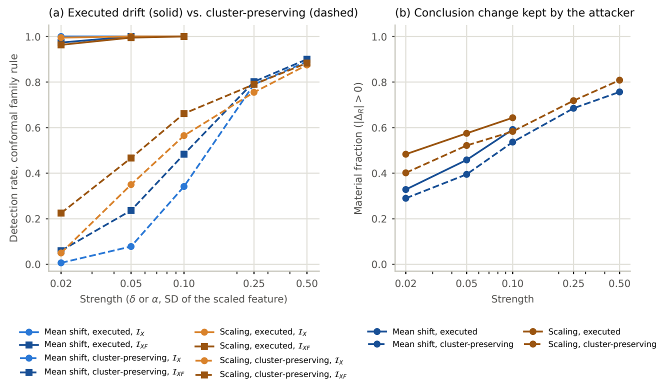

Fig. 2. Gate A. (a) Detection under the conformal family rule of the executed drift mechanisms (solid) and of their cluster-preserving variants (dashed) in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$, 600 observations per point. (b) Material fraction of the same rows: the attacker evades the batch-level sensors at matched strengths while keeping most of the conclusion changes.

the same cluster rule; the scaler sets how far the unclustered entries move. Of the 3,418 material adaptive rows, P2 serves 1,328 in $\mathcal{I}_X$, 959 in $\mathcal{I}_{XF}$ and 1,006 in $\mathcal{I}_{XFY}$, against 0, 6 and 10 of 1,848 for the executed mechanisms in the same runs; under $\mathcal{I}^{\star}_{XFY}$ every material adaptive row is blocked, because a changed conclusion with fixed labels requires a changed prediction. The "detected in every cell" result for feature drift was a property of the fingerprint, not of the mechanism: calibration controls false alarms but buys no coverage against an attacker who knows what the sensor sees; only a reference does.

6.5 Quantum-Workflow Coverage

All 165 cells are present and all nine prespecified acceptance checks pass; the supplement tabulates and plots the response fraction of every sensor per condition, read class by class. In this ideal-statevector gate the honest estimate equals the semantic kernel and g is the identity, so one anchored comparison against $\Phi(C_0)$ serves as the semantic probe on circuit-side rows and as the observed-kernel reference on post-processing rows; under finite-shot estimation the two anchors separate (Proposition 7(iii)). Clean cells trigger no sensor. Benign transpilation and the common-unitary rewrite change serialized provenance while preserving semantic kernels, algebraic properties and outputs in every cell (Proposition 7(i)): a raw hash difference needs policy-aware adjudication. Every data-dependent circuit mutation changes the semantic kernel; the mild RZ mutation changes no prediction, the stronger one changes predictions in 2/15 cells and the repetition change in 11/15. Asymmetric and diagonal-eroded matrices are detected algebraically in every cell and change no prediction; the PSD-preserving mixture is symmetric, unit-diagonal and PSD in all cells, so algebraic validation is exactly blind, the circuit hash sees nothing because C is unchanged, and only the kernel hash and the anchored comparison of the observed kernel expose all 15 substitutions, with output monitoring reacting in 1/15 (Proposition 7(ii)). Both shot emulators produce non-zero repeated-estimation discrepancy in all 30 cells and output changes in 7/15 and 3/15, a property of these cells rather than the universal law that Proposition 7(iii) declines to state; the contract holds them for adjudication instead of testing exact equality. The prototype's six envelopes verify and its eight acceptance checks pass (clean and approved transpilation allowed; parameterized mutation and PSD-preserving substitution blocked; prior-preserving label corruption blocked at the evaluation contract despite a zero class-prior delta; the approved 256-shot emulator held). The secondary ZZ-versus-SVC comparison (supplement) is retained only to show that the primary result does not depend on the model: its direction is stable, its magnitude conditional on scalers and clean headroom, and it is neither a quantum-advantage nor a vulnerability ordering.

7 Integrity-Audit Contract

The evidence yields a reporting and enforcement contract: every integrity claim names the boundary and protected asset, the adversary or failure class and what remains fixed, the information regime and trusted references it assumes, the sensor, the calibration rule and its decision-level level with its premise, the covered interventions, the structural, calibrated and adaptive residual blind regions, the cost on near-null variation, and the response. For the label path the minimum bundle is item-level label provenance plus a joint label-prediction audit; for the studied quantum boundary it is authenticated circuit/parameter provenance, an approved-transpilation policy, semantic probe kernels, a reference on the observed kernel, algebraic checks, repeated-estimation evidence and abstention where no null is calibrated. The contracts fail closed on their invariants (a violated invariant blocks, missing mandatory evidence holds); P2 may serve under an explicitly declared residual blind region; P3 abstains whenever a mandatory boundary lacks declared exact or statistical coverage and makes no promise about the power of the coverage that exists. Neither converts missing evidence into a claim of integrity; P2 converts it into a declared, accepted risk.

8 Discussion and Limitations

8.1 Security Interpretation

A metric can move while predictor output is bit-for-bit unchanged, and it can move upward; calling either model degradation assigns failure to the wrong component and suggests the wrong defence, since input robustness cannot repair corrupted ground truth while provenance and joint validation can. Zero sensor response is informative only if the sensor receives evidence capable of distinguishing the states, and non-zero separability only if the auditor holds a null tight enough to see it. The offline end-to-end result makes the consequence concrete: with the evidence a batch-level auditor normally has, about three fifths of the materially changed results in this suite would be served under a calibrated policy, and the share is set by the information regime, not by the threshold. The adversarial gate adds the second half of the lesson: a calibrated sensor family controls false alarms but not what an adaptive attacker can hide from it, so the "fully detected" mechanisms of the frozen suite are served in 28–39% of their material instances once the attacker preserves the fingerprint. The lever is provenance, whose granularity follows the integrity notion: a trusted aggregate of the same batch certifies the conclusion, item alignment certifies item identity, and nothing the adversary can rewrite certifies anything (Proposition 6). Its price is measured rather than hidden and is mostly statistical: the trusted policy interrupts 629 of 1,200 near-null synthetic controls, 544 statistical holds that the calibrated batch-level policy pays as well and 85 exact-reference blocks that the same-batch anchor adds.

8.2 Validity Boundaries

Three datasets and eight fixed scale/temporal settings, six of the nine environments from CICIDS2017, are not sampled deployments; five split clusters measure within-environment resampling only; most gates use 128/128 samples; balanced numeric-only staging does not reproduce operational prevalence. Both quantum evaluators are ideal statevector simulators with binomial shot emulation. SVC and QSVC keep pipeline-specific scalers, so their comparison does not isolate kernel geometry, and the same scalers govern how far an adaptive attacker's unclustered entries move. Exact blindness is a capability result under constructed perturbations and estimates no prevalence. The conformal level is marginal and holds under exchangeability of the calibration draws and the audited batch; the executed design violates that premise (draws overlap within a pool half, the calibration and evaluation halves are different rows of one table, and E1's pool half supplies one disjoint batch), the residual excess is reported as executed, and no conditional guarantee given a calibration set is claimed. The adversarial gate fixes one cluster rule and two mechanisms; an attacker with a different model of the sensors is not evaluated. The trusted item-aligned regime is a trust assumption, not a stronger detector: if the reference can be rewritten with the asset, Proposition 6 applies, and its interruption cost is measured on near-null synthetic controls, not on operational traffic. P3 abstains on missing coverage only; a boundary nominally covered by a low-power sensor remains a limitation of every policy here, and abstention conditioned on the validity of the calibration itself is not part of this layer. The prototype is a local process without authenticated provider records, and its hash chain certifies consistency of the record, not its authenticity. Every policy result is offline, on frozen outputs. Context-conditioned runtime calibration under non-stationarity, out-of-support abstention with recovery, real-QPU execution and the service-level evaluation of enforcement are the subject of a separate study.

9 Conclusion

Hybrid workflow integrity cannot be reduced to a robustness number. An intervention can change a reported conclusion without changing the model, a sensor can stay silent because its view is identical, a separable change can stay undetected because the auditor holds no same-batch anchor, and a detected change can become undetected once the attacker learns what the sensor sees. We formalized this as observational indistinguishability under explicit information sets and trusted references, derived monotone and exactly closable blind regions with the label path as a protected boundary, placed the quantum branch on the same lattice, confirmed the boundaries across three datasets and eight environments, calibrated the decision with a conformal rule whose finite-sample level holds under exchangeability and whose observed rates in the executed design are reported as such, and measured offline what each regime serves, what it costs on near-null variation and how an adaptive attacker evades it. Every integrity claim must therefore state its boundary, view, trusted roots, decision-level level and premise, residual and adaptive blind regions, cost and response; assurance is granted only when the evidence the claim requires is present.

Data, Code, and Ethics Statement

The review artifact contains source code, a locked environment, tests, compact derived tables, figures, seven SHA-256 evidence manifests, reproduction commands, the pre-registrations of the 1.1.0–1.3.0 gates and the amendment that replaced the calibration rule. Raw CICIDS2017, UNSW-NB15 and ToN-IoT data are not redistributed; their sources, expected hashes and deterministic staging are documented. The experimental evidence is frozen at artifact 1.3.0 (version DOI 10.5281/zenodo.22648573). Version 1.3.1 is archived at Zenodo, version DOI 10.5281/zenodo.22651111 (concept DOI 10.5281/zenodo.22550852); code is released under Apache-2.0, derived evidence, figures and documentation under CC BY 4.0. The study uses previously released benchmarks and involves no new human-participant or personal-data collection.

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